

|                     |                                |
|---------------------|--------------------------------|
| <b>Started on</b>   | Friday, 12 May 2023, 2:21 AM   |
| <b>State</b>        | Finished                       |
| <b>Completed on</b> | Saturday, 13 May 2023, 7:31 PM |
| <b>Time taken</b>   | 1 day 17 hours                 |
| <b>Grade</b>        | <b>94.75</b> out of 100.00     |

**Question 1**

Partially correct

Mark 23.75 out of 25.00

The figure below shows a randomly selected set of 10 samples of the Iris dataset we have introduced in the computer laboratory sessions. The numerical values of the original attributes of sepal length, sepal width, petal length and petal width have been transformed to binary values of long and short, respectively.

| sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | Species |
|-------------------|------------------|-------------------|------------------|---------|
| short             | short            | long              | long             | 1       |
| short             | long             | short             | short            | 0       |
| short             | short            | short             | long             | 1       |
| long              | long             | long              | long             | 2       |
| short             | long             | short             | short            | 0       |
| long              | short            | long              | long             | 2       |
| long              | short            | long              | long             | 1       |
| short             | long             | short             | short            | 0       |
| short             | long             | short             | short            | 0       |
| long              | long             | long              | long             | 1       |

Let SP, SL, SW, PL, PW represent Species, Sepal Length, Sepal Width, Petal Length, and Petal Width, respectively. Based on the 10 samples, fill in the Conditional Probability Tables below, respectively. Please use the modified formula we introduced in the lecture slides to avoid 0 probability, assuming  $m=1$ .

| P(SL   SP) | SP=0                                                                    | SP=1                                                                  | SP=2                                                                  |
|------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| SL='long'  |                                                                         |                                                                       |                                                                       |
|            | 1/10<br><br>Your last answer was interpreted as follows: $\frac{1}{10}$ | 1/2<br><br>Your last answer was interpreted as follows: $\frac{1}{2}$ | 5/6<br><br>Your last answer was interpreted as follows: $\frac{5}{6}$ |
| SL='short' |                                                                         |                                                                       |                                                                       |
|            | 9/10<br><br>Your last answer was interpreted as follows: $\frac{9}{10}$ | 1/2<br><br>Your last answer was interpreted as follows: $\frac{1}{2}$ | 1/6<br><br>Your last answer was interpreted as follows: $\frac{1}{6}$ |

| P(SW   SP) | SP=0                                                                    | SP=1                                                                    | SP=2                                                                  |
|------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|
| SW='long'  |                                                                         |                                                                         |                                                                       |
|            | 9/10<br><br>Your last answer was interpreted as follows: $\frac{9}{10}$ | 3/10<br><br>Your last answer was interpreted as follows: $\frac{3}{10}$ | 1/2<br><br>Your last answer was interpreted as follows: $\frac{1}{2}$ |

|            |                                                             |                                                             |                                                            |
|------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|
| SW='short' |                                                             |                                                             |                                                            |
|            | 1/10                                                        | 7/10                                                        | 1/2                                                        |
|            | Your last answer was interpreted as follows: $\frac{1}{10}$ | Your last answer was interpreted as follows: $\frac{7}{10}$ | Your last answer was interpreted as follows: $\frac{1}{2}$ |

|            |                                                             |                                                             |                                                            |
|------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|
| P(PL   SP) | SP=0                                                        | SP=1                                                        | SP=2                                                       |
| PL='long'  |                                                             | 7/10                                                        |                                                            |
|            | 1/10                                                        |                                                             | 3/6                                                        |
|            | Your last answer was interpreted as follows: $\frac{1}{10}$ | Your last answer was interpreted as follows: $\frac{7}{10}$ | Your last answer was interpreted as follows: $\frac{3}{6}$ |
| PL='short' |                                                             |                                                             |                                                            |
|            | 9/10                                                        | 3/10                                                        | 1/6                                                        |
|            | Your last answer was interpreted as follows: $\frac{9}{10}$ | Your last answer was interpreted as follows: $\frac{3}{10}$ | Your last answer was interpreted as follows: $\frac{1}{6}$ |

|            |                                                             |                                                             |                                                            |
|------------|-------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|
| P(PW   SP) | SP=0                                                        | SP=1                                                        | SP=2                                                       |
| PW='long'  |                                                             | 9/10                                                        | 5/6                                                        |
|            | 1/10                                                        |                                                             |                                                            |
|            | Your last answer was interpreted as follows: $\frac{1}{10}$ | Your last answer was interpreted as follows: $\frac{9}{10}$ | Your last answer was interpreted as follows: $\frac{5}{6}$ |
| PW='short' |                                                             |                                                             |                                                            |
|            | 9/10                                                        | 1/10                                                        | 1/6                                                        |
|            | Your last answer was interpreted as follows: $\frac{9}{10}$ | Your last answer was interpreted as follows: $\frac{1}{10}$ | Your last answer was interpreted as follows: $\frac{1}{6}$ |

Given a new sample where sepal length is long, sepal width is long, petal length is short, and petal width is short, using Naïve Bayes classifier to determine which species the new sample belongs to.

$P(\text{SP}=\text{'long'}, \text{SW}=\text{'long'}, \text{PL}=\text{'short'}, \text{PW}=\text{'short'} \mid \text{SP}=0) P(\text{SP}=0) =$

Your last answer was interpreted as follows:  $\frac{729}{25000}$

$P(\text{SP}=\text{'long'}, \text{SW}=\text{'long'}, \text{PL}=\text{'short'}, \text{PW}=\text{'short'} \mid \text{SP}=1) P(\text{SP}=1) =$

Your last answer was interpreted as follows:  $\frac{9}{5000}$

$P(\text{SP}=\text{'long'}, \text{SW}=\text{'long'}, \text{PL}=\text{'short'}, \text{PW}=\text{'short'} \mid \text{SP}=2) P(\text{SP}=2) =$

Your last answer was interpreted as follows:  $\frac{1}{432}$

So the new sample belongs to Species

Your last answer was interpreted as follows: 0

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A correct answer is  $\frac{1}{10}$ , which can be typed in as follows: 1/10

A correct answer is  $\frac{1}{2}$ , which can be typed in as follows: 1/2

A correct answer is  $\frac{5}{6}$ , which can be typed in as follows: 5/6

A correct answer is  $\frac{9}{10}$ , which can be typed in as follows: 9/10

A correct answer is  $\frac{1}{2}$ , which can be typed in as follows: 1/2

A correct answer is  $\frac{1}{6}$ , which can be typed in as follows: 1/6

A correct answer is  $\frac{9}{10}$ , which can be typed in as follows: 9/10

A correct answer is  $\frac{3}{10}$ , which can be typed in as follows: 3/10

A correct answer is  $\frac{1}{2}$ , which can be typed in as follows: 1/2

A correct answer is  $\frac{1}{10}$ , which can be typed in as follows: 1/10

A correct answer is  $\frac{7}{10}$ , which can be typed in as follows: 7/10

A correct answer is  $\frac{1}{2}$ , which can be typed in as follows: 1/2

A correct answer is  $\frac{1}{10}$ , which can be typed in as follows: 1/10

A correct answer is  $\frac{7}{10}$ , which can be typed in as follows: 7/10

A correct answer is  $\frac{5}{6}$ , which can be typed in as follows: 5/6

A correct answer is  $\frac{9}{10}$ , which can be typed in as follows: 9/10

A correct answer is  $\frac{3}{10}$ , which can be typed in as follows: 3/10

A correct answer is  $\frac{1}{6}$ , which can be typed in as follows: 1/6

A correct answer is  $\frac{1}{10}$ , which can be typed in as follows: 1/10

A correct answer is  $\frac{9}{10}$ , which can be typed in as follows: 9/10

A correct answer is  $\frac{5}{6}$ , which can be typed in as follows: 5/6

A correct answer is  $\frac{9}{10}$ , which can be typed in as follows: 9/10

A correct answer is  $\frac{1}{10}$ , which can be typed in as follows: 1/10

A correct answer is  $\frac{1}{6}$ , which can be typed in as follows: 1/6

A correct answer is  $\frac{729}{25000}$ , which can be typed in as follows: 729/25000

A correct answer is  $\frac{9}{5000}$ , which can be typed in as follows: 9/5000

A correct answer is  $\frac{1}{432}$ , which can be typed in as follows: 1/432

A correct answer is 0, which can be typed in as follows: 0

**Question 2**

Partially correct

Mark 21.00 out of 25.00

**Warning! Type out the operators in lower case (and/implies/or/not). Unfortunately 'iff' does not work in this question -- but you do not need it to answer any of the sub-questions.**

Convert the following expressions in a knowledge base into conjunctive normal form. Use proof by resolution to prove that it rained.

(1) If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.

(2) If the sailing race is held, then the trophy will be awarded.

(3) The trophy was not awarded.

Let  $R$ ,  $F$ ,  $S$ ,  $L$ , and  $T$  represent the sentences 'it rains', 'it is foggy', 'sailing race is held', 'lifesaving demonstration will go on', and 'the trophy will be awarded', respectively.

The conjunction normal form (CNF) are:

(1)  $(R \text{ or } S) \text{ and } (F \text{ or } S) \text{ and } (R \text{ c$

Your last answer was interpreted as follows:  $(R \text{ or } S) \text{ and } (F \text{ or } S) \text{ and } (R \text{ or } L) \text{ and } (F \text{ or } L)$

The variables found in your answer were:  $[F, L, R, S]$

(2)  $\text{not } S \text{ or } T$

Your last answer was interpreted as follows:  $\text{not } (S) \text{ or } T$

The variables found in your answer were:  $[S, T]$

(3)  $\text{not } T$

Your last answer was interpreted as follows:  $\text{not } (T)$

The variables found in your answer were:  $[T]$

Negating the conclusion: (4)  $\text{not } R$

Your last answer was interpreted as follows:  $\text{not } (R)$

The variables found in your answer were:  $[R]$

In the following, prove it rained by resolution algorithms in 3 steps.

Based on cnf (1) and cnf (2), we have: (5)  $R \text{ or } T$

Your last answer was interpreted as follows:  $R \text{ or } T$

The variables found in your answer were:  $[R, T]$

Based on cnf (5) and cnf (3), we have: (6)  $R$

Your last answer was interpreted as follows:  $R$

The variables found in your answer were:  $[R]$

Based on cnf (6) and cnf (4), we have: (7)  $R \text{ or not } R$

Your last answer was interpreted as follows:  $R \text{ or not } (R)$

The variables found in your answer were:  $[R]$

A correct answer is  $(R \text{ or } S) \text{ and } (F \text{ or } S) \text{ and } (R \text{ or } L) \text{ and } (F \text{ or } L)$ , which can be typed in as follows:  $(R \text{ or } S) \text{ and } (F \text{ or } S) \text{ and } (R \text{ or } L) \text{ and } (F \text{ or } L)$

A correct answer is not  $(S)$  or  $T$ , which can be typed in as follows: not  $S$  or  $T$

A correct answer is not  $(T)$ , which can be typed in as follows: not  $T$

A correct answer is not  $(R)$ , which can be typed in as follows: not  $R$

A correct answer is  $R$  or  $T$ , which can be typed in as follows:  $R$  or  $T$

A correct answer is  $R$ , which can be typed in as follows:  $R$

A correct answer is  $R$  and not  $(R)$ , which can be typed in as follows:  $R$  and not  $R$

**Question 3**

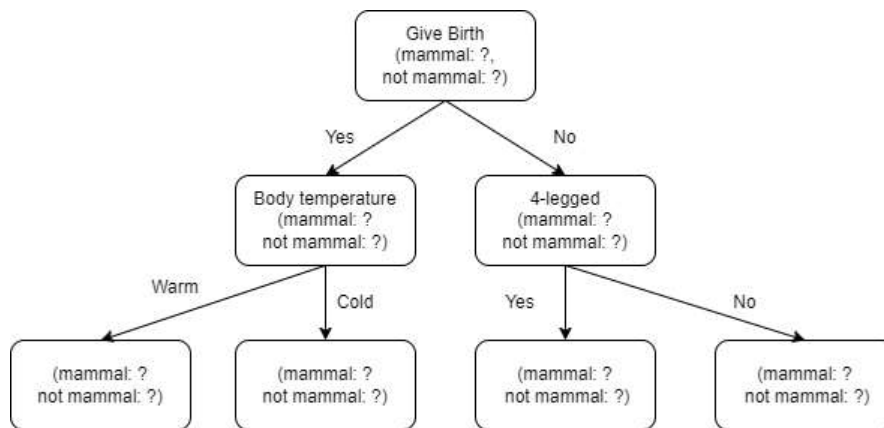
Correct

Mark 25.00 out of 25.00

The table below contains a sample of records of animals.

| Name            | Body Temperature | Given Birth | 4-legged | Hibernates | Class Label (mammal) |
|-----------------|------------------|-------------|----------|------------|----------------------|
| Human           | Warm Blooded     | Y           | N        | N          | Yes                  |
| Pigeon          | Warm Blooded     | N           | N        | N          | No                   |
| Elephant        | Warm Blooded     | Y           | Y        | N          | Yes                  |
| Leo pant Shark  | Cold Blooded     | Y           | N        | N          | No                   |
| Turtle          | Cold Blooded     | N           | Y        | N          | No                   |
| Penguin         | Cold Blooded     | N           | N        | N          | No                   |
| Fish Anya       | Cold Blooded     | N           | N        | N          | No                   |
| Dolphin         | Warm Blooded     | Y           | N        | N          | Yes                  |
| Spiny Nocturnal | Warm Blooded     | N           | Y        | Y          | Yes                  |
| Gia Frog        | Cold Blooded     | N           | Y        | Y          | No                   |

Using the attributes *Given Birth* as the root node, *Body Temperature* as the left child of the root node, and *4-legged* as the right child of the root node, we can draw a decision tree of depth 2 as follows to classify the 10 samples based on the attribute *Class Label* (mammal).



For each node, show the number of samples in each class.

For the root node, *Give Birth*, the number of mammals is

Your last answer was interpreted as follows: 4

, and the number of non-mammals is

Your last answer was interpreted as follows: 6

For the node, *Body Temperature*, the number of mammals is

Your last answer was interpreted as follows: 3

, and the number of non-mammals is

Your last answer was interpreted as follows: 1

For the node, *4-legged*, the number of mammals is

Your last answer was interpreted as follows: 1

, and the number of non-mammals is

Your last answer was interpreted as follows: 5

For the *left child* node of *Body Temperature*, the number of mammals is

Your last answer was interpreted as follows: 3

, and the number of non-mammals is

Your last answer was interpreted as follows: 0

For the *right child* node of *Body Temperature*, the number of mammals is

Your last answer was interpreted as follows: 0

, and the number of non-mammals is

Your last answer was interpreted as follows: 1

For the *left child* node of *4-legged*, the number of mammals is

Your last answer was interpreted as follows: 1

, and the number of non-mammals is

Your last answer was interpreted as follows: 2

For the *right child* node of *4-legged*, the number of mammals is

Your last answer was interpreted as follows: 0

, and the number of non-mammals is

Your last answer was interpreted as follows: 3

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A correct answer is 4, which can be typed in as follows: 4

A correct answer is 6, which can be typed in as follows: 6

A correct answer is 3, which can be typed in as follows: 3

A correct answer is 1, which can be typed in as follows: 1

A correct answer is 1, which can be typed in as follows: 1

A correct answer is 5, which can be typed in as follows: 5

A correct answer is 3, which can be typed in as follows: 3

A correct answer is 0, which can be typed in as follows: 0

A correct answer is 0, which can be typed in as follows: 0

A correct answer is 1, which can be typed in as follows: 1

A correct answer is 1, which can be typed in as follows: 1

A correct answer is 2, which can be typed in as follows: 2

A correct answer is 0, which can be typed in as follows: 0

A correct answer is 3, which can be typed in as follows: 3

## Question 4

Correct

Mark 25.00 out of 25.00

The table below contains a set of 10 randomly selected samples of the fruit dataset on Kaggle.

| fruit_label | fruit_name | fruit_subtype    | mass | width | height | color_score |
|-------------|------------|------------------|------|-------|--------|-------------|
| 1           | apple      | cripps_pink      | 162  | 7.4   | 7.2    | 0.85        |
| 1           | apple      | braeburn         | 166  | 6.9   | 7.3    | 0.93        |
| 3           | orange     | turkey_navel     | 154  | 7.2   | 7.2    | 0.82        |
| 1           | apple      | cripps_pink      | 160  | 7.5   | 7.5    | 0.86        |
| 1           | apple      | braeburn         | 154  | 7.0   | 7.1    | 0.88        |
| 3           | orange     | turkey_navel     | 190  | 7.5   | 8.1    | 0.74        |
| 4           | lemon      | unknown          | 132  | 5.8   | 8.7    | 0.73        |
| 1           | apple      | golden_delicious | 168  | 7.5   | 7.6    | 0.73        |
| 2           | mandarin   | mandarin         | 76   | 5.8   | 4.0    | 0.81        |
| 1           | apple      | braeburn         | 178  | 7.1   | 7.8    | 0.92        |

(1) If we use the attributes *mass*, *width*, *height*, and *color\_score*, to predict the attribute *fruit\_label*, it is a problem of

Your last answer was interpreted as follows: 1

(2) If we use the attribute *color\_score*, to predict the attribute *fruit\_name*, it is a problem of

Your last answer was interpreted as follows: 1

(3) If we use the attributes *mass*, *width*, and *height*, to predict *color\_score*, it is a problem of

Your last answer was interpreted as follows: 2

(4) Suppose you will train a linear regression model to predict the *height* of a fruit from its *width*, write down the objective function based on the sample data from the last 5 rows of the table. Please use the linear regression model  $y = wx + b$  where  $x$  is the input variable and  $y$  is the output variable. Use \* for multiplication and ^ for power. **Please do not use square brackets [ ].**

The objective function is:

Your last answer was interpreted as follows:

$$(7.5 \cdot w + b - 7.6)^2 + (7.5 \cdot w + b - 8.1)^2 + (7.1 \cdot w + b - 7.8)^2 + (5.8 \cdot w + b - 4.0)^2 + (5.8 \cdot w + b - 8.7)^2$$

The variables found in your answer were:  $[b, w]$

(5) Assume there are in total 60 samples in the dataset. You are required to use a 5-fold cross validation to train and evaluate the linear regression model in (4).

How many times in total will you train the model?

Your last answer was interpreted as follows: 5

There are

Your last answer was interpreted as follows: 48

number of samples in a training set and

Your last answer was interpreted as follows: 12

number of samples in a validation set, respectively, each time you train the model.

How would you report the validation performance of the model you trained?

average MSE

Your last answer was interpreted as follows: 1

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A correct answer is: "classification"

A correct answer is: "classification"

A correct answer is: "regression"

A correct answer is

$(7.5 \cdot w + b - 7.6)^2 + (7.5 \cdot w + b - 8.1)^2 + (7.1 \cdot w + b - 7.8)^2 + (5.8 \cdot w + b - 4.0)^2 + (5.8 \cdot w + b - 8.7)^2$ , which can be typed in as follows:  $(7.5*w+b-7.6)^2+(7.5*w+b-8.1)^2+(7.1*w+b-7.8)^2+(5.8*w+b-4.0)^2+(5.8*w+b-8.7)^2$

A correct answer is 5, which can be typed in as follows: 5

A correct answer is 48, which can be typed in as follows: 48

A correct answer is 12, which can be typed in as follows: 12

A correct answer is: "average MSE"